Proof

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

#k-combinations of $\{1,2,3,\ldots,n\}$

#$(k-1)$-combinations of $\{1,2,3, \ldots, n-1\}$ containing last element

#k-combinations of $\{1,2,3,\ldots, n-1\}$ not containing last element

Use the Sum Rule for the total number of combinations